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BSIT 2-2N

INFORMATION MANAGEMENT ASSESSMENT

PART 1

Answer the following questions

1. Give your own examples (3 examples) of functional dependencies between attributes.
Use line with arrow to show the determinant and the dependent attributes

Example 1 (Student Table):

StudentID → StudentName

StudentID → Course

Example 2 (Employee Table):

EmployeeID → EmployeeName

EmployeeID → Department

Example 3 (Product Table):

ProductID → ProductName

ProductID → Price

2. Give your own example of record with multi-valued attributes. Show all attribute names and give data for each attribute highlighting the multi-valued attributes

Example: Student Record

StudentID: 2025001

Name: Aaron Celindro

Course: BSIT

PhoneNumbers: {09123456789, 09987654321} ← (multi-valued)

Skills: {Python, SQL, JavaScript} ← (multi-valued)

PhoneNumbers and Skills have multiple values

3. Give other examples (3) of attributes which can be derived (any record).

Example Record (Employee):

Birthdate → Age

BasicSalary → AnnualSalary

HoursWorked × HourlyRate → TotalPay

So:

Age is derived from Birthdate

AnnualSalary = BasicSalary × 12

TotalPay = HoursWorked × Rate

4. Explain how domain constraints impose restrictions on data to be input into the attribute/table/database

Domain constraints ensure that only valid and appropriate data is entered into a database by restricting the type, format, or range of values allowed in each attribute.

Example:

Age must be between 0–120

Gender must be 'Male' or 'Female'

Email must follow email format

5. Explain how normalization can remove redundancy

Normalization reduces data redundancy in a database by decomposing large, cluttered tables into smaller, related tables and defining relationships between them. This process ensures that each piece of data is stored in only one place, which eliminates unintentional duplication, saves storage space, and prevents data anomalies.

Example

student_data			
<u>StudentID</u>	<u>Name</u>	<u>Course</u>	<u>Instructor</u>
1	Aaron	IT	Sir John
2	Mark	IT	Sir John

Note: "Sir John" is repeated

After Normalization:

- Student Table
- Course Table
- Instructor Table

NO MORE REPEATED DATA